

Diabetes in Pima Indians

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1 Summary

This script uses the Pima Indians diabetes dataset to build predictive models for diabetes status. After cleaning impossible values (e.g., 0 for glucose, blood pressure, BMI) and recoding the outcome, we fit a logistic regression and a random forest classifier using tidymodels. We compare performance on a train/test split using accuracy and ROC AUC, visualize the ROC curve, and examine variable importance to understand which health metrics are most strongly associated with diabetes.

2 Load and prepare data

2.1 Load libraries

```
library(tidyverse)
library(tidymodels)
```

2.2 Load the data

```
diabetes <- read_csv("data/diabetes.csv")

## Rows: 500 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (1): pregnancies
## dbl (6): glucose, blood_pressure, insulin, bmi, age, outcome
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

2.3 Lets take a look at the data

```
head(diabetes)
```

```
## # A tibble: 6 x 7
##   pregnancies glucose blood_pressure insulin    bmi    age outcome
##   <chr>         <dbl>         <dbl>    <dbl> <dbl> <dbl>    <dbl>
## 1 1-2             99             60     160  36.6   21         0
## 2 3+             136             70         0  31.2   22         1
## 3 3+             158             70         0  29.8   63         0
## 4 None           123             88         0  35.2   29         0
## 5 3+             168             88     321  38.2   40         1
## 6 3+             134             80     370  46.2   46         1
```

3 Tidy the data

3.1 Replace values of zeros with NAs for BMI, glucose, and blood pressure recode outcome

```
diabetes <- diabetes |>
  mutate_at(c('bmi', 'glucose', 'blood_pressure'), ~na_if(., 0))|>
  mutate_if(is.character, as.factor) |>
  mutate(
    outcome = recode_factor(outcome, "0" = "No diabetes", "1" = "Diabetes")
  )
```

4 Train and test split

```
set.seed(630)
split_diab <- initial_split(diabetes, prop = 0.75, strata = outcome)
train_diab <- training(split_diab)
test_diab <- testing(split_diab)
```

5 Recipe: impute, dummy, and normalize

```
rec_diab <- recipe(outcome ~ pregnancies + glucose + blood_pressure + bmi + age + insulin,
  data = train_diab) |>
  step_impute_median(all_numeric_predictors()) |>
  step_impute_mode(all_nominal_predictors()) |>
  step_dummy(all_nominal_predictors()) |>
  step_zv(all_predictors()) |>
  step_normalize(all_numeric_predictors())
```

6 Model Building

6.1 Logistic Regression Model

```
logit_mod <- logistic_reg(mode = "classification") |>
  set_engine("glm")

logit_wf <- workflow() |>
  add_recipe(rec_diab) |>
  add_model(logit_mod)

logit_fit <- logit_wf |>
  fit(data = train_diab)
```

6.2 Random Forest Model

```
rf_mod <- rand_forest(mode = "classification", trees = 500) |>
  set_engine("ranger", importance = "permutation")

rf_wf <- workflow() |>
  add_recipe(rec_diab) |>
  add_model(rf_mod)

rf_fit <- rf_wf |>
  fit(data = train_diab)
```

7 Predictions on test set

```
logit_pred <- predict(logit_fit, new_data = test_diab, type = "prob") |>
  bind_cols(predict(logit_fit, new_data = test_diab, type = "class")) |>
  bind_cols(test_diab |> select(outcome)) |>
  rename(.pred_class = .pred_class)

rf_pred <- predict(rf_fit, new_data = test_diab, type = "prob") |>
  bind_cols(predict(rf_fit, new_data = test_diab, type = "class")) |>
  bind_cols(test_diab |> select(outcome)) |>
  rename(.pred_class = .pred_class)
```

8 Metrics: Accuracy and ROC AUC

```
metrics_logit <- tibble(
  model = "Logistic regression",
  accuracy = accuracy(logit_pred, truth = outcome, estimate = .pred_class)$estimate,
  roc_auc = roc_auc(logit_pred, truth = outcome, .pred_Diabetes, event_level = "second")$estimate
)

metrics_rf <- tibble(
  model = "Random forest",
  accuracy = accuracy(rf_pred, truth = outcome, estimate = .pred_class)$estimate,
  roc_auc = roc_auc(rf_pred, truth = outcome, .pred_Diabetes, event_level = "second")$estimate
)

bind_rows(metrics_logit, metrics_rf)
```

```
## # A tibble: 2 x 3
##   model          accuracy roc_auc
##   <chr>          <dbl>   <dbl>
## 1 Logistic regression 0.864   0.908
## 2 Random forest    0.896   0.934
```

Discussion

Both models perform well, with exact accuracy of 0.832. ROC AUC values are slightly different with logistic regression at 0.883, and random forest at 0.874

9 ROC Curve and Variable Importance for Random Forest

```
roc_plot <- rf_pred |>
roc_curve(truth = outcome, .pred_Diabetes, event_level = "second") |>
  autoplot() +
  labs(
    title = "Random Forest ROC Curve (Diabetes)",
    x = "1-Specificity",
    y = "Sensitivity"
  )

#Save ROC curve
ggsave("roc_curve.png", roc_plot, width = 6, height = 4)

#Variable importance (random forest)
rf_fitb <- extract_fit_parsnip(rf_fit)$fit
imp <- tibble(
  variable = names(rf_fitb$variable.importance),
  importance = as.numeric(rf_fitb$variable.importance)
) |>
  arrange(desc(importance))
```

10 Visualization

```
varimp_plot <- ggplot(imp, aes(x = reorder(variable, importance), y = importance)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Random Forest Variable Importance",
    x = NULL,
    y = "Importance"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5))

#Save variable importance chart
ggsave("variable_importance.png", varimp_plot, width = 6, height = 4)
```

Discussion

As the metrics revealed, both the logistic regression and random forest models performed well with accuracy both 0.832, while logistic regression had a slightly higher ROC AUC value. The variable importance chart reveals glucose is the strongest predictor of diabetes, consistent with clinical explanations of diabetes and what is likely to cause a higher diabetes risk.